# Optimising Business Performance through Data Management and Analytics

# A Case Study on Sampat Kumar & Co

A Proposal report for the BDM Capstone Project

Submitted by

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# Declaration Statement

I am working on a project titled “Optimising Business performance through data management and analysis – A Case Study on Sampat Kumar and Co” . I extend my appreciation to Sampat Kumar and Co Shop for providing the necessary resources that enabled me to conduct my project.

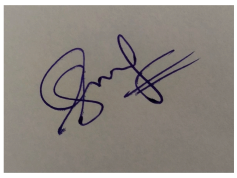
I hereby assert that the data presented and assessed in this project report is genuine and precise to the utmost extent of my knowledge and capabilities. The data has been gathered from primary sources and carefully analysed to ensure its reliability.

Additionally, I affirm that all procedures employed for the purpose of data collection and analysis have been duly explained in this report. The outcomes and inferences derived from the data are an accurate depiction of the findings acquired through thorough analytical procedures.

I am dedicated to adhering to the principles of academic honesty and integrity, and I am receptive to any additional examination or validation of the data contained in this project report.

I understand that the execution of this project is intended for individual completion and is not to be undertaken collectively. I thus affirm that I am not engaged in any form of collaboration with other individuals, and that all the work undertaken has been solely conducted by me. In the event that plagiarism is detected in the report at any stage of the project's completion, I am fully aware and prepared to accept disciplinary measures imposed by the relevant authority.

I understand that all recommendations made in this project report are within the context of the academic project undertaken towards course fulfilment in the BS Degree Program offered by IIT Madras. The institution does not endorse any of the claims or comments.



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# 1.Executive Summary

Sampat Kumar & Co. is a well-known family-run wholesale and retail store located in Visakhapatnam. It deals in food and packaging products and has been serving both business clients and regular customers for decades. Despite having a good reputation, the store is now facing some real challenges in managing its daily operations efficiently.

The main issues include keeping too much inventory that sometimes expires, a shortage of storage space, and slow billing during rush hours. These problems affect both the staff and customers, making it harder to grow the business. In this project, I plan to use simple tools like Excel and Google Sheets to collect and analyze data from the shop. I will look at stock usage patterns, how space is used in the shop, and how long it takes to bill customers. Based on this, I'll suggest practical solutions like improving inventory planning, rethinking storage setup, and exploring faster billing options. The goal is to make daily operations smoother, reduce losses, and give customers a better shopping experience.

# 2. Organisation background

Sampat Kumar & Co. was started in 1962 and is located in Poorna Market, Visakhapatnam. It is a trusted name in the local wholesale and retail space, dealing in a variety of products like Basmati rice, sweets, bakery items, and packaging materials. The business serves both individual customers and other businesses. Currently, there are 8 employees, and the shop operates daily from 9 AM to 10:30 PM. Monthly revenue averages between ₹15 to ₹20 lakhs, and the shop sees around 100–150 customers a day.

# 3. Problem Statement

### 3.1 Inventory Control and Profit Leakage

The absence of proper inventory rotation and tracking systems causes items—especially

perishables—to go unsold or expire. This ties up capital in unsellable stock and reduces the store's profit margins.

### 3.2 Ineffective Use of Floor Space

Limited physical space, combined with unstructured layout practices, leads to disorganized storage. This slows down restocking and product access, impacting daily efficiency and limiting sales capacity.

### 3.3 Customer Wait Time and Billing Bottlenecks

Manually handling customer billing during peak hours results in long queues and delayed service. This not only frustrates customers but may cause some to abandon purchases, reducing overall daily revenue."

## 4. Background of the Problem

Sampat Kumar & Co. is currently facing a combination of internal inefficiencies and external constraints that limit its operational effectiveness. Internally, the business relies heavily on manual processes such as handwritten inventory logs and billing through calculators. Externally, the increase in customer footfall and the limited shop floor space put pressure on daily operations.

One key issue is inefficient inventory management. Without a real-time inventory tracking system, the shop struggles to maintain accurate stock levels. This results in both overstocking and stockout. Overstocking leads to expired goods, particularly perishables such as liquid food items, while stockout during high-demand periods can result in missed sales. These inefficiencies tie up working capital and reduce profitability.

The shop also suffers from poor utilisation of its limited floor space. Storage areas lack proper categorisation and vertical organisation, making it difficult for staff to locate or restock items quickly. This not only slows down operations but also limits the store's ability to expand its product range. The disorganised layout increases clutter, reducing customer convenience and staff productivity.

Billing is another major bottleneck. During peak hours, the absence of digital billing systems results in long queues and delayed service. Customers may abandon purchases due to frustration, which directly affects daily revenue. Moreover, manual billing introduces errors, making it harder to track sales performance accurately. These challenges highlight the urgent need for data-driven interventions to streamline operations, improve space usage, and enhance the overall customer experience.

## 5. Problem Solving Approach

To address these challenges, a structured and data-driven strategy will be implemented focusing on inventory control, space optimization, and billing efficiency. The approach uses quantitative techniques and affordable digital tools such as Microsoft Excel, Google Sheets, and simple POS simulations to ensure results are measurable and actionable.

**Inventory Management Optimization:** A FIFO (First-In, First-Out) approach will be implemented using Excel to track purchase and expiry dates. Conditional formatting will highlight aging inventory, reducing spoilage. Additionally, ABC analysis will categorize stock based on sales frequency and value (A – high value/fast-moving; B – moderate; C – low value/ slow-moving). Alerts for low or expiring stock will improve replenishment timing. KPI: Reduction in expired inventory and improved stock turnover.

**Space Utilization and Layout Planning:** A digital map of the shop layout will be created and analyzed to optimize product placement. Time-motion studies will be conducted to measure how long it takes employees to access key products. Post-intervention, shelf rearrangement and vertical stacking will be introduced to improve space use. KPI: Decrease in average product retrieval time and increase in items per square foot.

**Billing Process Optimization:** Billing speed will be measured during peak hours using stopwatch logs. A barcode-based billing simulation will be introduced using mock setups in Excel. The comparison between manual and simulated digital billing will assess improvements. KPI: Reduction in average billing time and drop in customer abandonment rate.

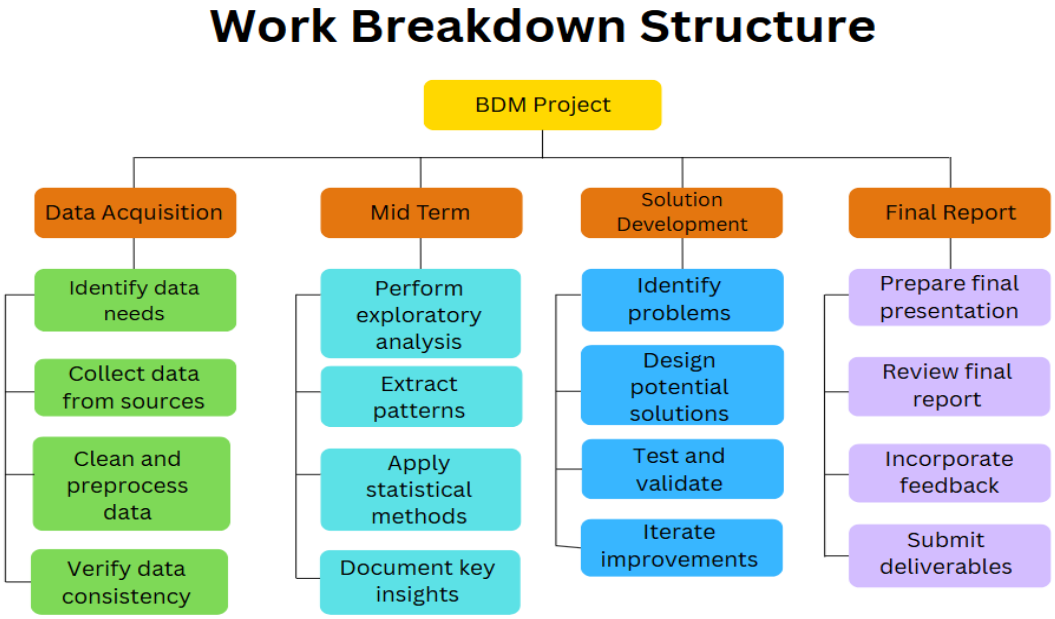
**Qualitative Improvements and KPI Quantification:** Improvements in customer and employee satisfaction will be tracked through simple feedback forms. The number of complaints, perceived wait time, and staff-reported ease of access to inventory will be used to evaluate the success of changes. KPI: Reduction in customer complaints and qualitative feedback trends.

**Data Sources and Tools:** Stock registers, billing logs, and observation records will be used for data collection. Excel and Google Sheets will be the primary tools for analysis, visualization, and simulations. These tools provide an effective and low-cost method to perform quantitative analysis and support operational decision-making.

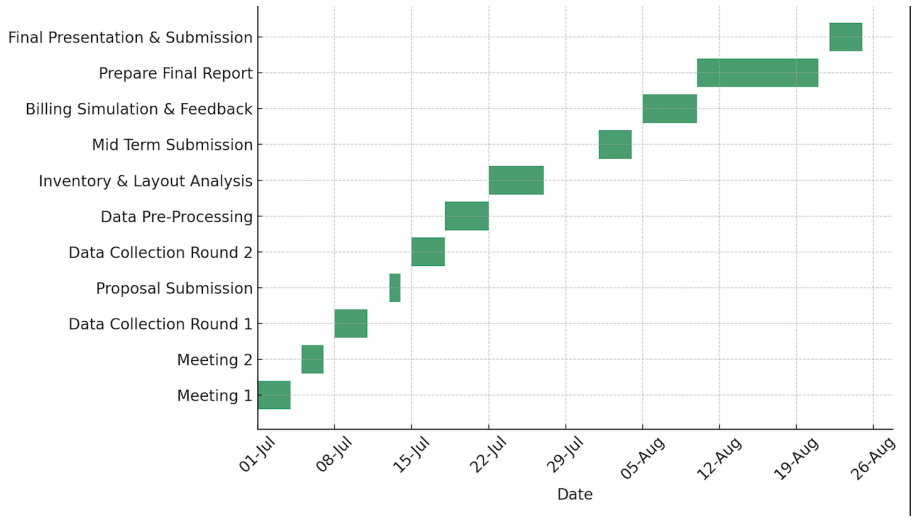
By applying these solutions, Sampat Kumar & Co. aims to reduce losses, improve efficiency, and enhance customer service, thereby enabling sustainable business growth.

# 6 Expected Timeline

### 6.1 Work Breakdown Structure:



### 6.2 Gantt Chart:




# 7.Expected Outcome

### 7.1 Inventory Efficiency

Expired or unsold items will reduce significantly through the FIFO method and ABC analysis. This will improve product availability and ensure that capital is not tied up in dead stock. A measurable drop in inventory waste is expected.

### 7.2 Store Space Utilization

The new layout and shelving strategy will make better use of the available floor and vertical space. Staff will be able to access items faster, and the store can display a greater range of products. Time taken for product retrieval is expected to decrease.

### 7.3 Customer Service Improvement

Billing times during peak hours will be shorter, which will reduce long queues and improve customer experience. Customer satisfaction is expected to improve through faster service, reduced complaints, and more accurate billing.